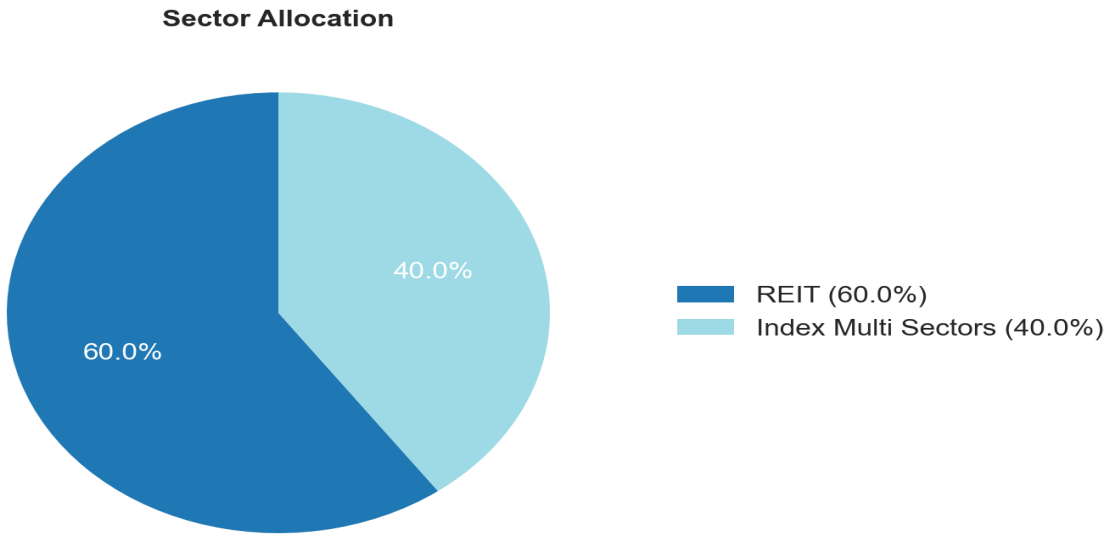


Portfolio rotation report

Date of extraction	2025-12-06 14:43
Data window	2019-12-30 → 2024-10-21
Frequency	Weekly — 252 observations. Sufficient depth: short- and long-term indicators are statistically reliable. Warning: the observation window exceeds the typical interpretability range for this frequency (+384.6% above the recommended upper bound). Results may blend multiple market regimes, reducing economic signal clarity.
Tickers	IYR, QQQ, SPY, VNQ, XLRE
Indicators	Short term: 8 weeks volatility, 12 weeks return, 16 weeks momentum Long term: 28 weeks volatility, 30 weeks return, 80 weeks momentum

1. Portfolio Composition

- XLRE (Real Estate Select Sector SPDR Fund — REIT)
 - SPY (SPDR S&P; 500 ETF — Index Multi Sectors)
- VNQ (Vanguard Real Estate ETF — REIT)
 - IYR (iShares US Real Estate ETF — REIT)
 - QQQ (Invesco QQQ Trust — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Highly concentrated structure: limited diversification and sensitivity to one macro driver.
- Broad balance across market capitalizations: diversification achieved across the size spectrum.

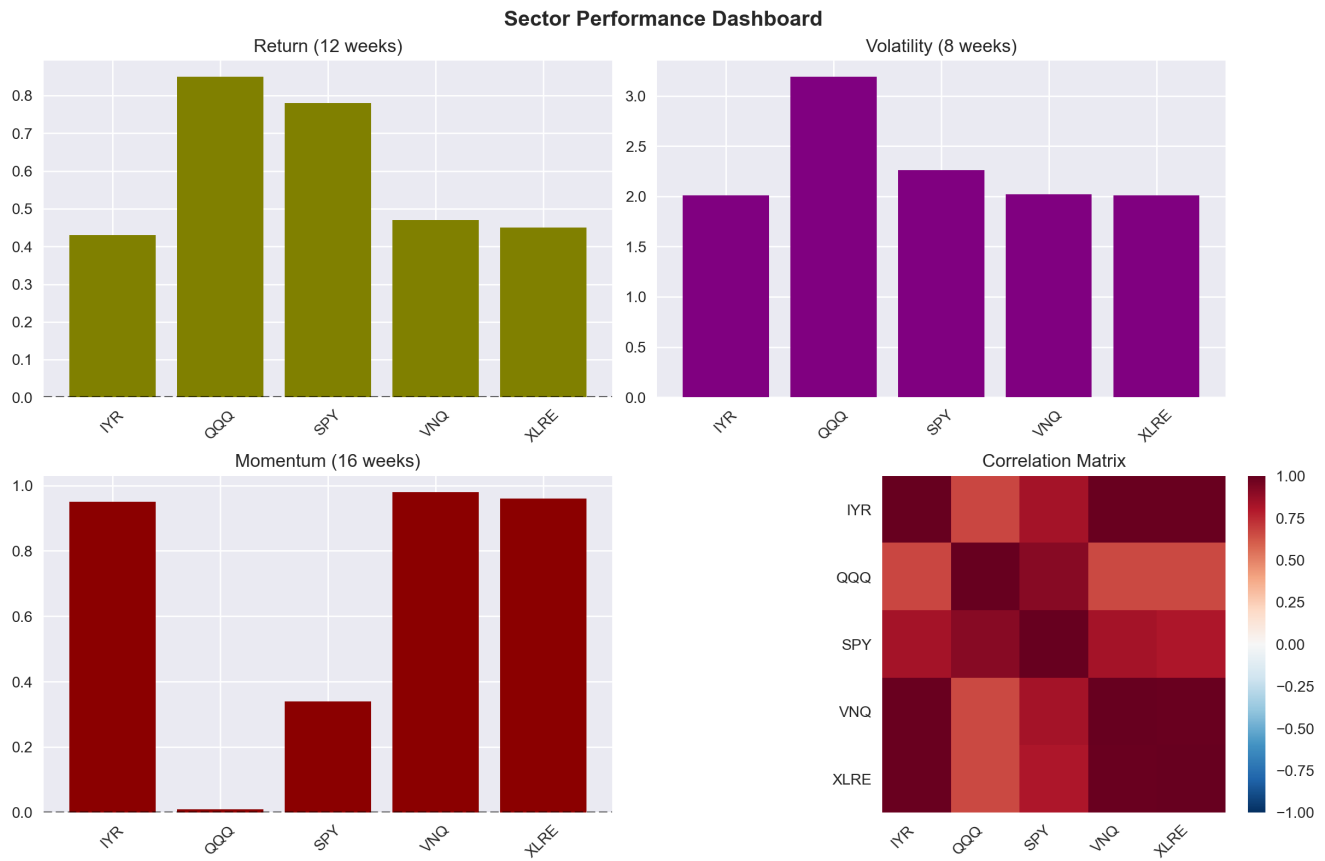
Structural properties

- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names.
- Defensive beta tilt: lower systematic risk profile.
- Balanced volatility structure: risk evenly distributed across holdings.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations
100.0% of portfolio

Pair	Correlation
IYR / XLRE	0.99
IYR / VNQ	0.99
VNQ / XLRE	0.99
QQQ / SPY	0.91
SPY / VNQ	0.83

Overall, average cross-sector correlation was 0.83, ranging from 0.66 to 0.99. This suggests very high correlation — sectors behave almost as a single macro block. Correlation dispersion is narrow, suggesting similar risk dynamics across sectors. Strongest co-movement between IYR and VNQ (0.99); weakest between QQQ and VNQ (0.66). Market-driven regime: macro factors dominate individual sector dynamics.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
IYR	0.43	2.01	0.216	0.95	mildly coherent	subdued	thin	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.
QQQ	0.85	3.19	0.266	0.01	neutral	subdued	liquid	mixed short-term dynamics, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
SPY	0.78	2.26	0.346	0.34	mildly coherent	subdued	liquid	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, with ample liquidity.
VNQ	0.47	2.02	0.232	0.98	mildly coherent	subdued	illiquid	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.
XLRE	0.45	2.01	0.226	0.96	mildly coherent	subdued	thin	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
QQQ	0.85
SPY	0.78
VNQ	0.47
XLRE	0.45
IYR	0.43

Top by Volatility

Ticker	Volatility(%)
QQQ	3.19
SPY	2.26
VNQ	2.02
IYR	2.01
XLRE	2.01

Top by Momentum

Ticker	Momentum(%)
VNQ	0.98
XLRE	0.96
IYR	0.95
SPY	0.34
QQQ	0.01

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
IYR	0.05	2.98	0.239	1.02	0.57	6.90 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.
QQQ	0.51	0.01	0.01	1.11	1.18	53.05 M	stable performance, momentum fading, volatility normalizing, aligned with market dynamics, under robust trading volume.
SPY	0.41	0.61	0.211	1.24	1.00	72.91 M	stable performance, momentum fading, volatility expanding, aligned with market dynamics, under robust trading volume.
VNQ	0.08	3.07	0.241	1.04	0.59	3.60 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.
XLRE	0.07	2.92	0.245	1.01	0.56	5.90 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.

■ Light-impact observations

For IYR, QQQ, SPY, VNQ, XLRE: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.6%, momentum 0.6%, vol 2.3%)